MS1Vtx_rms_clusE_raw

MC vs data · overview · rank 32 / 98


```
rank 32 / 98 by endcap D_equal

used by: NN1 barrel, NN1 endcap

rank-identical columns merged into this one:
    MS1Vtx_barrel_rms_clusE
    MS1Vtx_endcap_rms_clusE
```

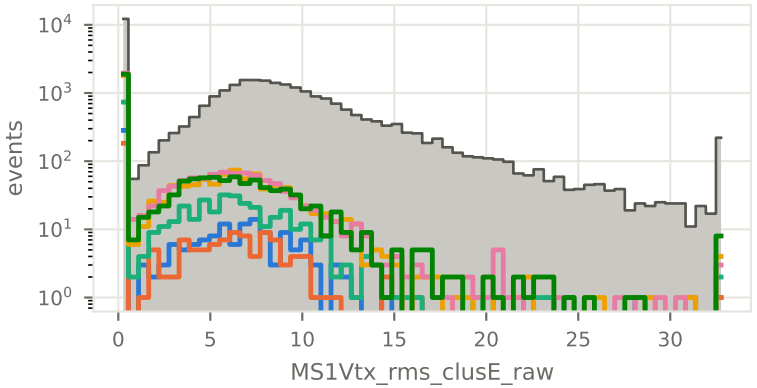
region	D_equal	noise	D_pooled	125/15	125/55	200/50	400/100	600/150	1000/275	n_bkg	rank
endcap	0.365	0.022	0.372	0.362	0.342	0.372	0.368	0.373	0.384	34,970	32/98
barrel	0.299	0.051	0.309	0.331	0.261	0.324	0.322	0.317	0.298	1,089	36/98
all	0.413	0.018	0.420	0.405	0.399	0.418	0.414	0.424	0.428	36,059	19/98

MS1Vtx_rms_clusE_raw

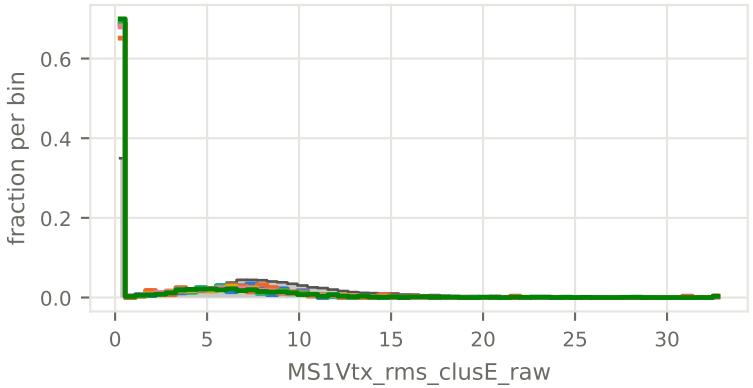
region: ENDCAP · rank 32 / 98 · in: NN1 barrel, NN1 endcap

D_equal 0.365 (noise 0.022) D_pooled 0.372 min 0.342 (mH125_mS55) max 0.384 (mH1000_mS275) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

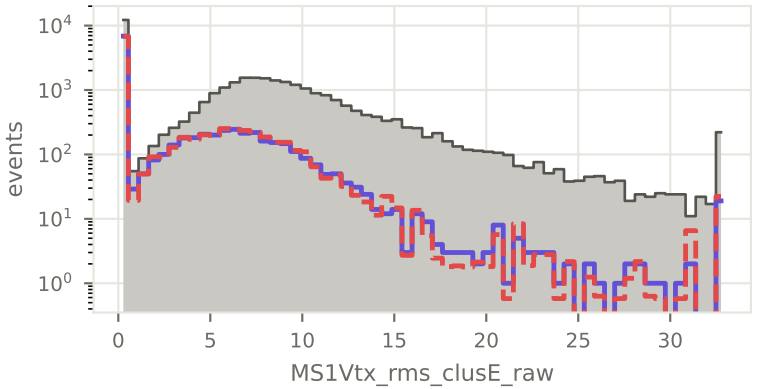


1b · each mass point vs background · normalized

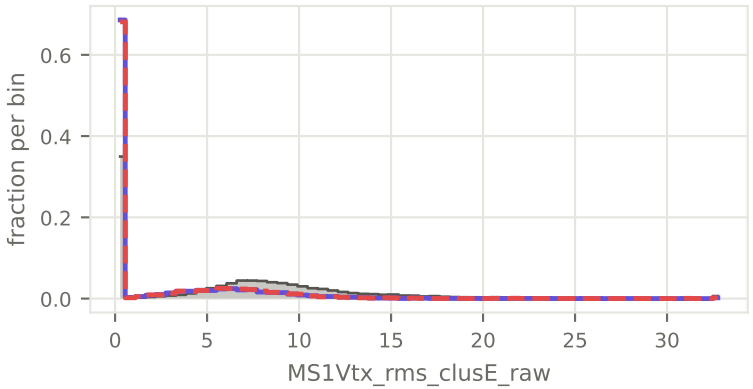


1 · each mass point
background n 34,970
mH125_mS15 D 0.362 n 411
mH125_mS55 D 0.342 n 281
mH200_mS50 D 0.372 n 1,071
mH400_mS100 D 0.368 n 2,648
mH600_mS150 D 0.373 n 2,882
mH1000_mS275 D 0.384 n 2,703

2a · combined signal vs background · raw

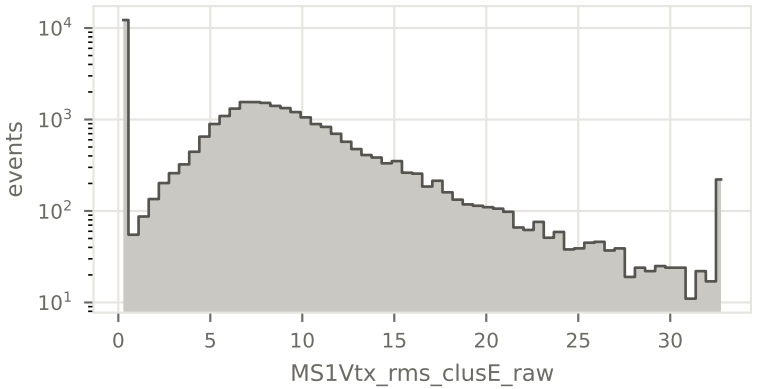


2b · combined signal vs background · normalized

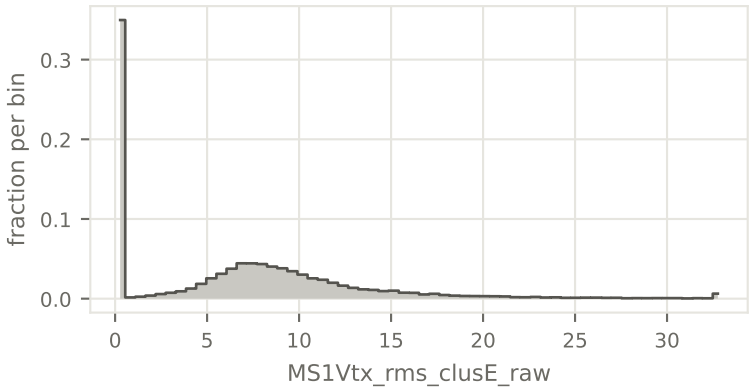


2 · combined signal
background n 34,970
combined, pooled D 0.372 n 9,996
equal weight per mass D 0.365

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

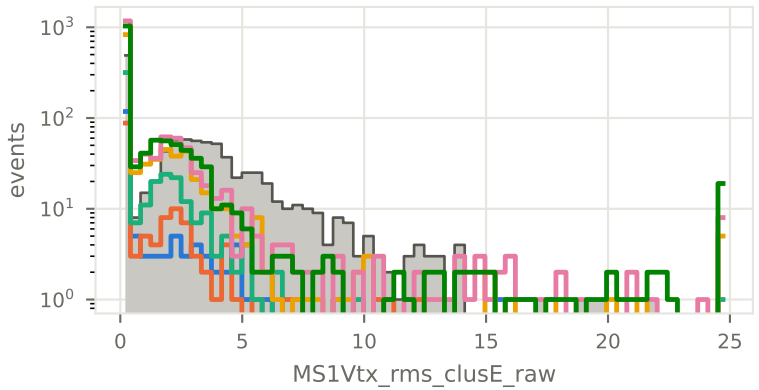
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

MS1Vtx_rms_clusE_raw

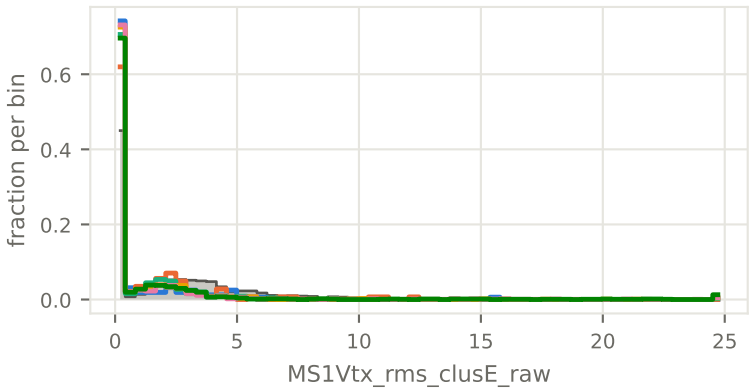
region: BARREL · rank 32 / 98 · in: NN1 barrel, NN1 endcap

D_equal 0.299 (noise 0.051) D_pooled 0.309 min 0.261 (mH125_mS55) max 0.331 (mH125_mS15) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

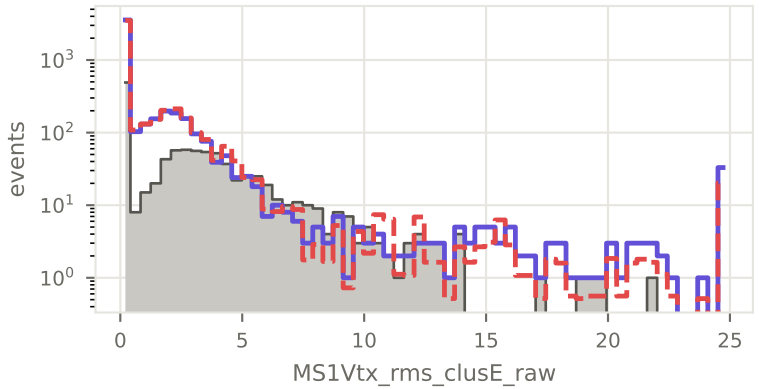


1b · each mass point vs background · normalized

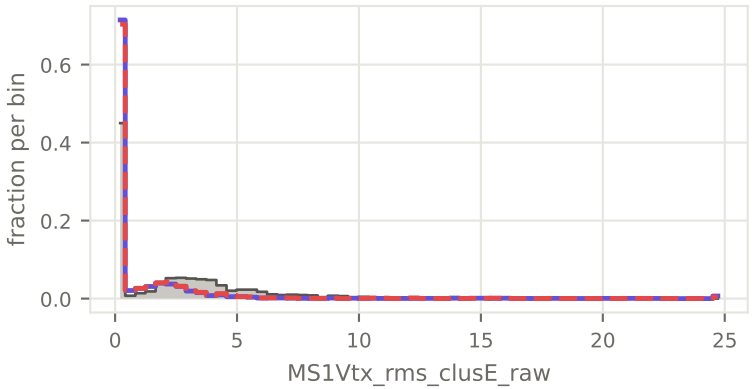


1 · each mass point
background n 1,089
mH125_mS15 D 0.331 n 159
mH125_mS55 D 0.261 n 142
mH200_mS50 D 0.324 n 449
mH400_mS100 D 0.322 n 1,144
mH600_mS150 D 0.317 n 1,608
mH1000_mS275 D 0.298 n 1,485

2a · combined signal vs background · raw

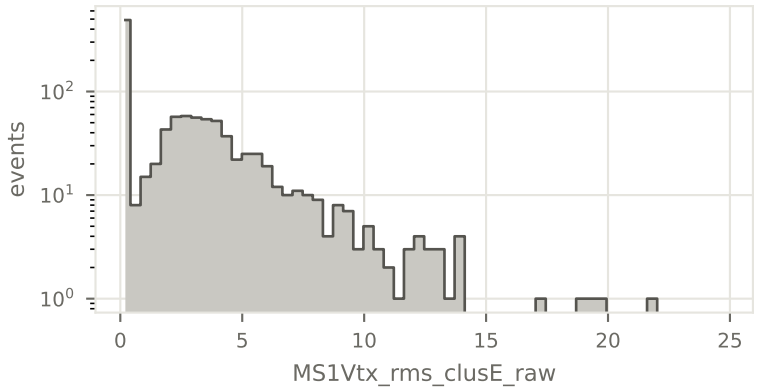


2b · combined signal vs background · normalized

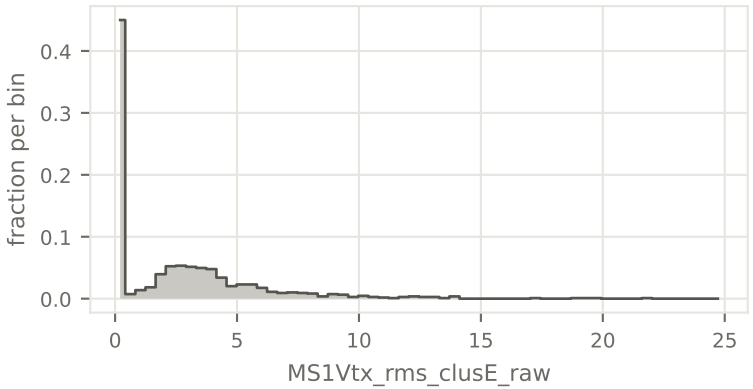


2 · combined signal
background n 1,089
combined, pooled D 0.309 n 4,987
equal weight per mass D 0.299

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

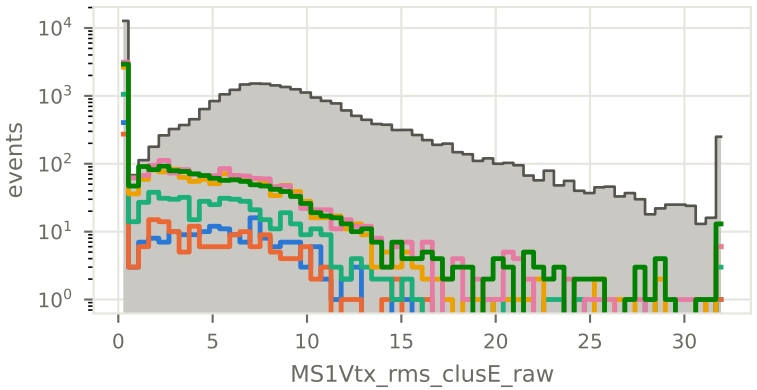
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

MS1Vtx_rms_clusE_raw

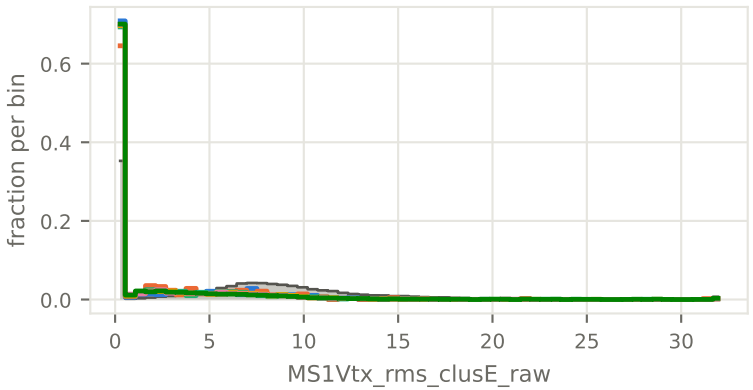
region: ALL · rank 32 / 98 · in: NN1 barrel, NN1 endcap

D_equal 0.413 (noise 0.018) D_pooled 0.420 min 0.399 (mH125_mS55) max 0.428 (mH1000_mS275) bkg n = 36,059 sig n = 14,983

1a · each mass point vs background · raw

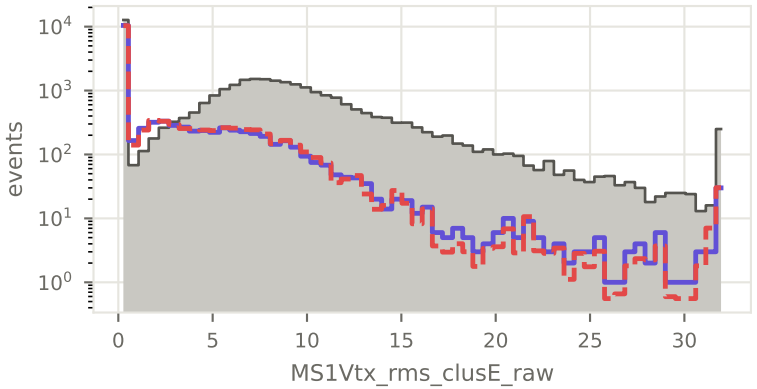


1b · each mass point vs background · normalized

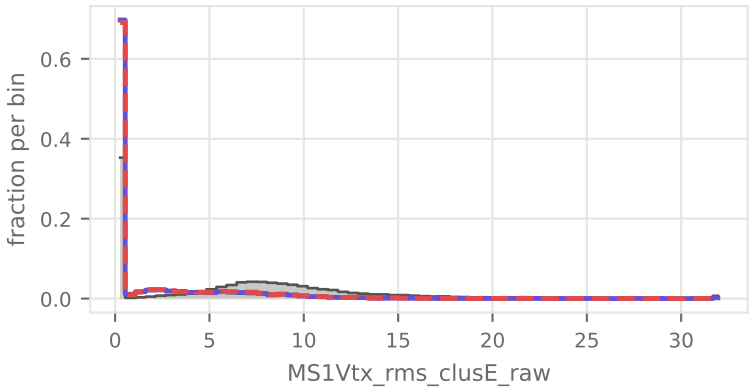


1 · each mass point
- background n 36,059
- mH125_mS15 D 0.405 n 570
- mH125_mS55 D 0.399 n 423
- mH200_mS50 D 0.418 n 1,520
- mH400_mS100 D 0.414 n 3,792
- mH600_mS150 D 0.424 n 4,490
- mH1000_mS275 D 0.428 n 4,188

2a · combined signal vs background · raw

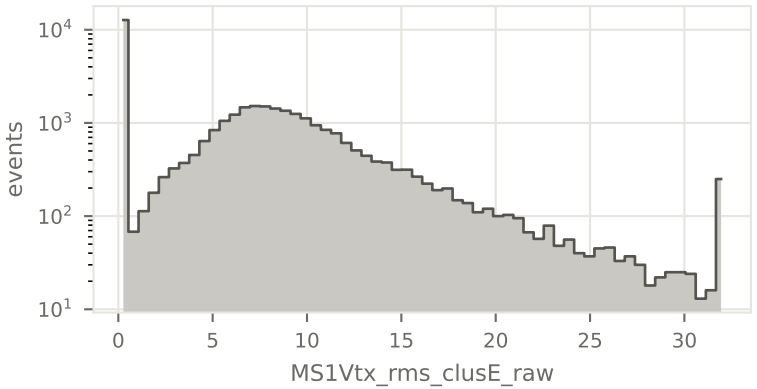


2b · combined signal vs background · normalized

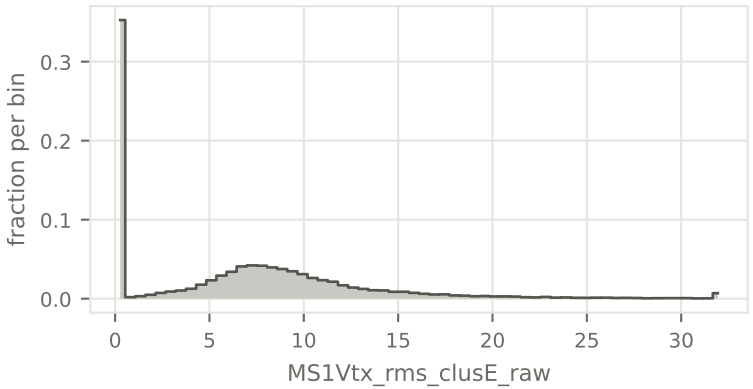


2 · combined signal
- background n 36,059
- combined, pooled D 0.420 n 14,983
- equal weight per mass D 0.413

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
- background n 36,059

a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed